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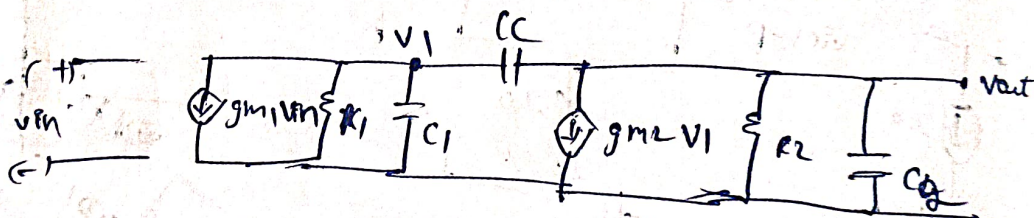
$$Z_1 \Rightarrow \frac{g_{m2}}{C_C}$$

$$\frac{V_O}{V_{in}}$$

$$A_{DC} \left(1 - \frac{s}{Z_1} \right)$$

$$A_{DC} \Rightarrow g_{m1} R_1 \cdot g_{m2} R_2$$

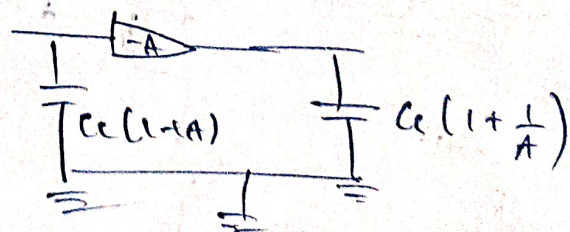
$$\left(1 + \frac{s}{P_1} \right) \left(1 + \frac{s}{P_2} \right)$$



$$\frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_1} \times \frac{V_1}{V_{in}}$$

$$\left(\frac{1}{sC_1} \right) + \frac{V_1}{R_1} + g_{m1} V_{in} + \frac{V_1 - V_{out}}{\frac{1}{sC_C}} = 0$$

$$V_1 = \frac{V_O \cdot sC_C R_1 - g_{m1} R_1 V_{in}}{1 + sR_1(C_1 + C_2)}$$



Miller capacitances

$$\left(\frac{1}{sC_2} \right) + \frac{V_{out}}{R_2} + g_{m2} V_{in} + \frac{V_{out} - V_{in}}{\left(\frac{1}{sC_c} \right)} = 0$$

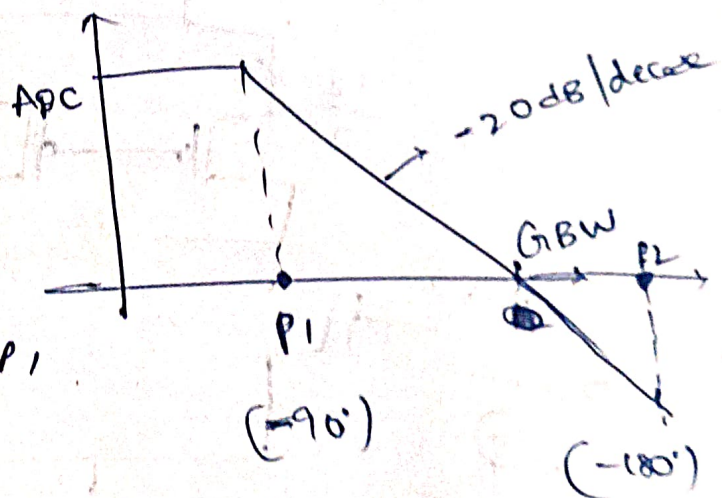
$$V_{out} \left[s(C_2 + C_c) + \frac{1}{R_2} \right] = V_{in} (sC_c - g_{m2})$$

Miller Capacitances cannot be used because of

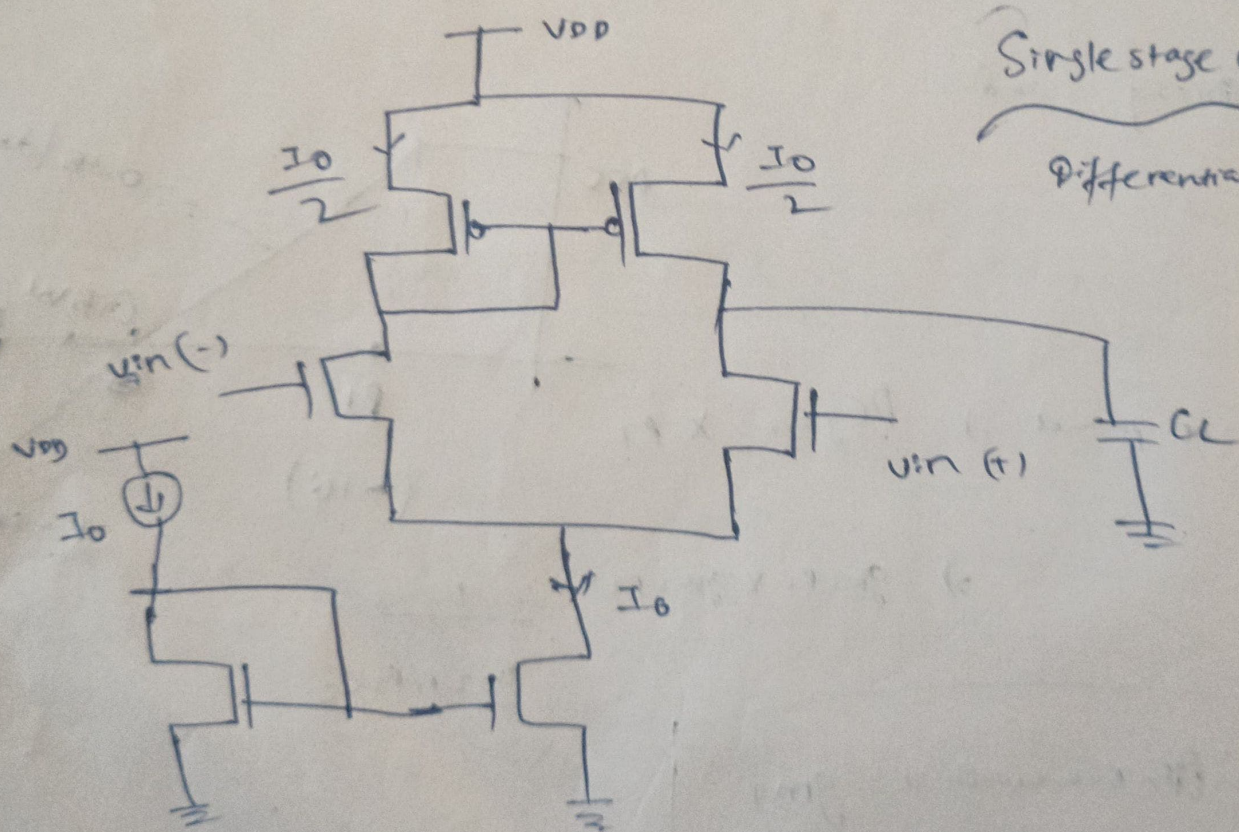
- 1) pole splitting and
- 2) zero.

Gain Bandwidth product $\Rightarrow$ $DC \text{ gain} \times P_1$

$$\Rightarrow g_{m1} R_1 \times g_{m2} R_2 \times \frac{1}{g_{m2} R_2 C_c}$$

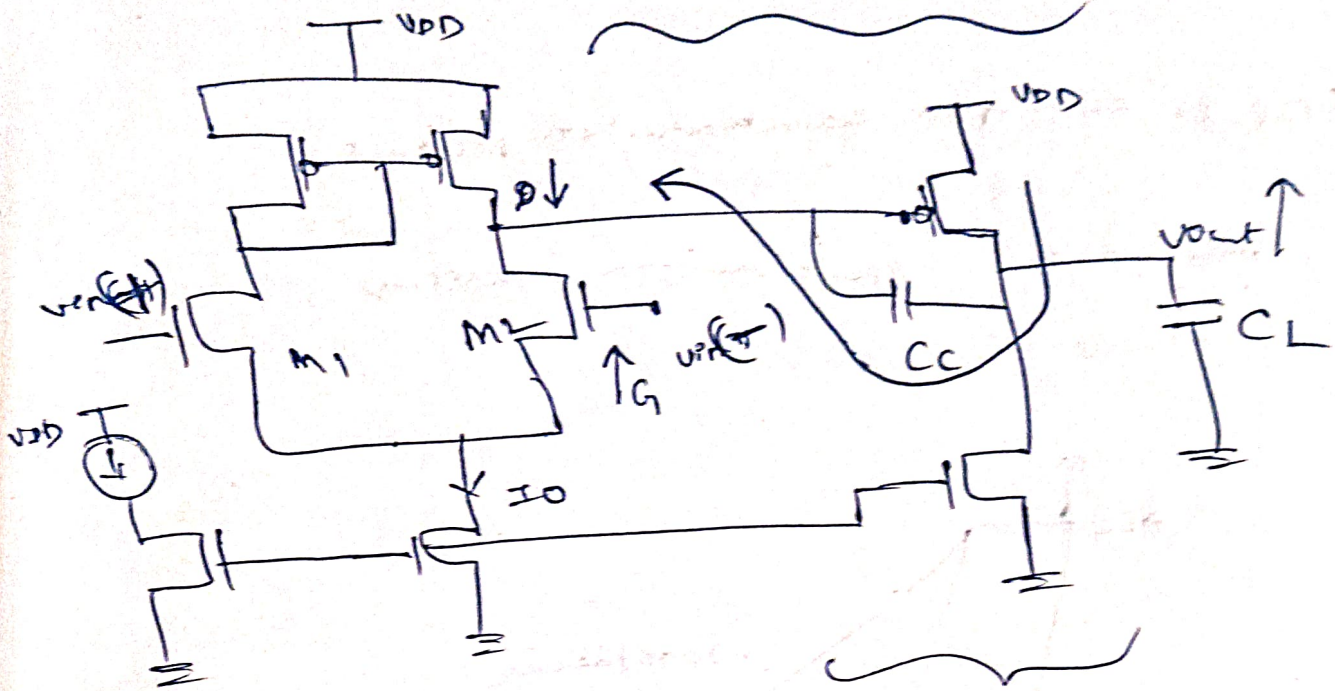


$$\text{Gain Bandwidth product} \Rightarrow \frac{g_{m1}}{C_c}$$



Single stage opamp
Differential Amplifier.

Two stage opamp

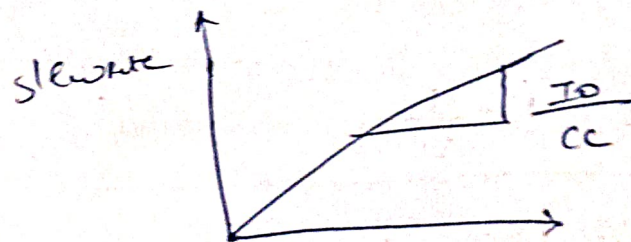


Diff Amp

Cs Amp

Slewrate

Rate of change of output



*
$$\text{Slewrate} \Rightarrow \frac{I_O}{C_c}$$

Assume $v_{in}(+)$ is $\odot$

current goes through

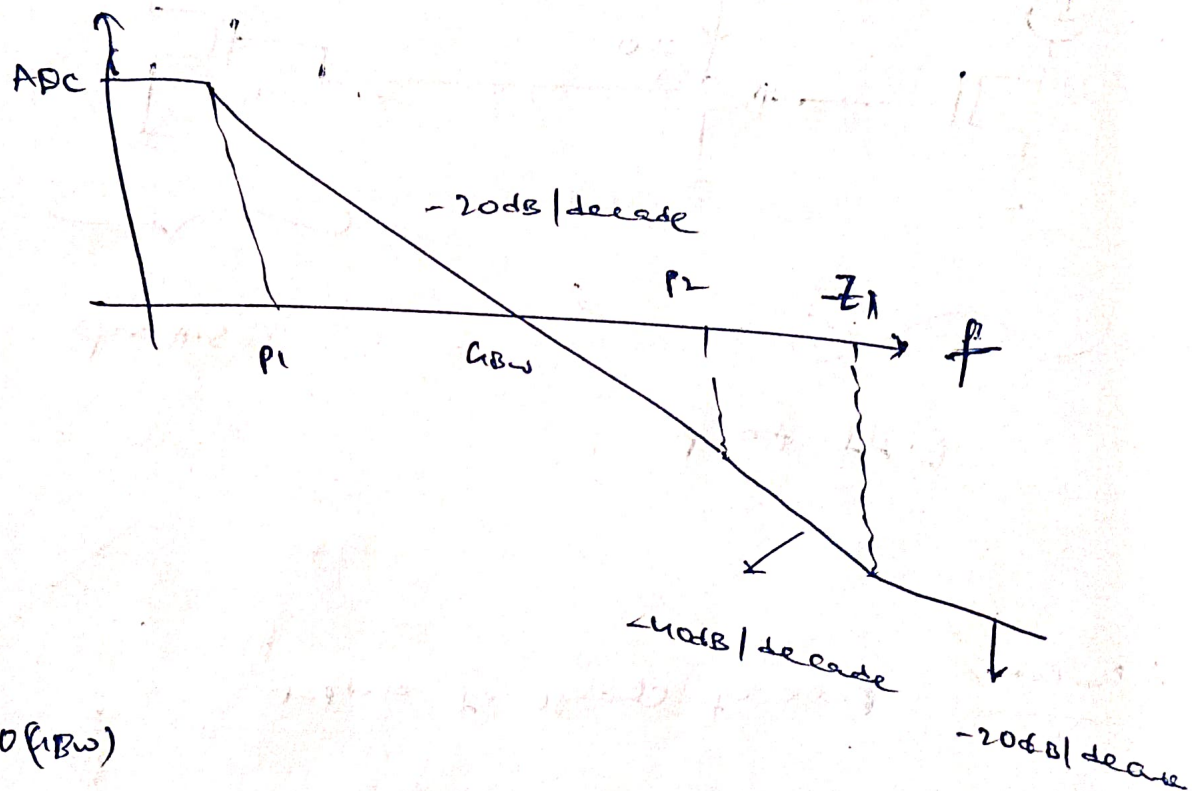
for M_2 ON.
 M_1 OFF.

$v_{in}(-)$
Transistor.

Phase margin

Keep p_2 away from gain bandwidth product:

otherwise phase margin decreases



Assume

$$z_1 \geq 10(\text{GBW})$$

$$\frac{V_O}{V_{IN}} \approx \frac{AD_c \left(1 - \frac{s}{z_1} \right)}{\left(1 + \frac{s}{p_1} \right) \left(1 + \frac{s}{p_2} \right)}$$

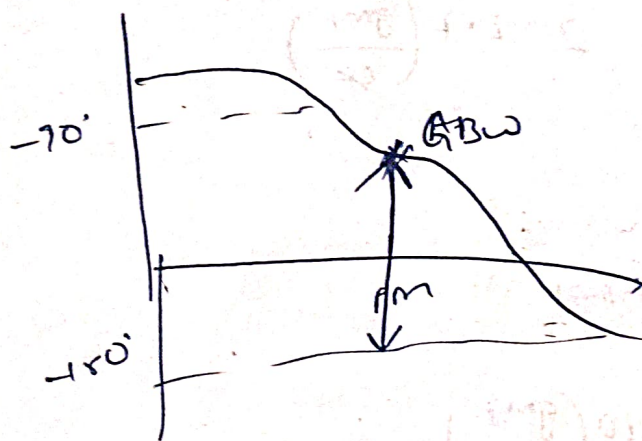
$$\angle \frac{V_O}{V_{IN}} \approx -\tan^{-1} \left(\frac{\omega}{z_1} \right) - \tan^{-1} \left(\frac{\omega}{p_1} \right) - \tan^{-1} \left(\frac{\omega}{p_2} \right)$$

At $\omega = \omega_{GBW}$

$$\left(\frac{V_O}{V_{in}} \right) = -\tan^{-1} \left(\frac{A_{GBW}}{z_1} \right) - \tan^{-1} \left(\frac{A_{GBW}}{p_1} \right) = \tan^{-1} \left(\frac{A_{GBW}}{p_2} \right)$$

Substitute $z_1 = \frac{gm_2}{CC} = 10 A_{GBW}$ $A_{GBW} = \frac{gm_1}{CC}$

$p_1 = \frac{1}{gm_1 R_1 CC}$ $p_2 = \frac{gm_2}{C_2}$



$$-180 + PM = -\tan^{-1} \left(\frac{1}{10} \right) - \tan^{-1} \left(\frac{A_{GBW}}{p_2} \right) - \tan^{-1} \left(\frac{A_{GBW}}{p_2} \right)$$

* for $PM = 60^\circ \rightarrow \frac{A_{GBW}}{p_2} = 0.4513$

Ex

for $PM \geq 60^\circ$ then

$p_2 \geq 2.2 A_{GBW}$